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Paper Code: TBT-101

(B. Tech Biotechnology)
Mid Semester Examination September-2025
I Semester

Paper Name: FUNDAMENTALS OF MEDICAL BIOTECHNOLOGY

Time: 1:30 Hours

MM: 50

Note:

- (i) All questions are compulsory.
- (ii) Answer any ONE sub questions among a, & b in each main question.
- (iii) Total marks for each main question are ten.

Q1.

- (a) Define Infection? With suitable examples of each, discuss different types of infections? **CO1**
- (b) Describe bacterial capsule? With examples, comment on its composition and their significance in bacterial pathogenicity. **CO1**

Q2

- (a.) Comment on the general characteristics of viruses? Using diagrams, describe the morphology of the following:

- i. Rhabdovirus
- ii. Human Immunodeficiency virus

CO1

- (b.) Define Mycoses? With examples, discuss the general characteristics, and laboratory diagnosis of medically important fungi. **CO2**

Q3.

- a. Describe the structure and function of bacterial flagella. Using specific examples, discuss about the types of bacterial flagella. **CO2**
- b. Differentiate between Gram- positive and Gram-negative bacterial cell wall? Give two examples of each Gram-positive bacilli and Gram-negative bacilli shaped bacteria. **CO1**

Q4.

- a. Differentiate between commensal and transient microflora? Comment on the natural microflora of skin and gastrointestinal tract? **CO1**
- b. Using diagram, discuss any "One" method of cultivation of viruses in laboratory. **CO2**

Q5.

- a. Describe the morphology, virulence factors associated with *Clostridium tetani*? **CO1**
- b. Discuss in brief the pathogenicity, and clinical manifestation of *Bacillus cerus*? **CO1**

